

FIRST SEMESTER (C.B.C.S.S.) DEGREE EXAMINATION, NOVEMBER 2020**Mathematics****MTS 1B 01—BASIC LOGIC AND NUMBER THEORY****(2020 Admissions)****(Multiple Choice Questions for SDE Candidates)****Time : 15 Minutes****Total No. of Questions : 20****Maximum : 20 Marks****INSTRUCTIONS TO THE CANDIDATE**

1. This Question Paper carries Multiple Choice Questions from 1 to 20.
2. The candidate should check that the question paper supplied to him/her contains all the 20 questions in serial order.
3. Each question is provided with choices (A), (B), (C) and (D) having one correct answer. Choose the correct answer and enter it in the main answer-book.
4. The MCQ question paper will be supplied after the completion of the descriptive examination.

MTS 1B 01—BASIC LOGIC AND NUMBER THEORY

(Multiple Choice Questions for SDE Candidates)

1. 'What is the value of x after this statement, assuming initial value of x is 5? 'If x equals to one then $x = x + 2$ else $x = 0$ ' :
 - (A) 1.
 - (B) 3.
 - (C) 0.
 - (D) 2.
2. The compound statement $A \rightarrow (A \rightarrow B)$ is false, then the truth values of A, B are respectively.
 - (A) T, T.
 - (B) F, T.
 - (C) T, F.
 - (D) F, F.
3. Let $P(x)$ denote the statement " $x > 7$ ". Which of these have truth value true ?
 - (A) $P(0)$
 - (B) $P(4)$.
 - (C) $P(6)$.
 - (D) $P(9)$.
4. When to proof $P \rightarrow Q$ true, we proof P false, that type of proof is known as :
 - (A) Direct proof.
 - (B) Contrapositive proofs.
 - (C) Vacuous proof.
 - (D) Mathematical Induction.
5. In the principle of mathematical induction, which of the following steps is mandatory ?
 - (A) Induction hypothesis.
 - (B) Inductive reference.
 - (C) Induction set assumption.
 - (D) Minimal set representation.
6. By induction hypothesis, the series $1^2 + 2^2 + 3^2 + \dots + p^2$ can be proved equivalent to :
 - (A) $p^2 + 27$.
 - (B) $\frac{p * (p+1) * (2p+1)}{6}$.
 - (C) $\frac{p * (p+1)}{4}$.
 - (D) $p + p^2$.
7. For any positive integer m , ... is divisible by 4.
 - (A) $5m^2 + 2$.
 - (B) $3m + 1$.
 - (C) $m^2 + 3$.
 - (D) $m^3 + 3m$.

8. Let $\gcd(a, b) = d$. If c divides a and c divides b , then :
- (A) $c \leq d$. (B) $c \geq d$.
 (C) $c = 1$. (D) Cannot determine c with the given information.
9. If $a \mid b$ and $a \mid c$, then :
- (A) $b \mid c$. (B) $c \mid a$.
 (C) $a \mid (b + c)$. (D) $b \mid a$.
10. If a and b are non zero integers with $a \mid b$, then $\gcd(a, b)$ equals :
- (A) $|a|$. (B) b .
 (C) ab . (D) a .
11. The product of any three consecutive integers is divisible by :
- (A) 36. (B) 9.
 (C) 6. (D) 8.
12. If a is an odd integer then $\gcd(3a, 3a + 2)$ equals :
- (A) 3. (B) 5.
 (C) 1. (D) 2.
13. $(1001111)_2 = \underline{\hspace{2cm}}$.
- (A) 79. (B) 89.
 (C) 69. (D) 99.
14. The linear Diophantine equation $ax + by = c$ has a solution if and only if :
- (A) $\gcd(a, c) \mid b$. (B) $\gcd(a, b) \mid c$.
 (C) $\gcd(c, b) \mid a$. (D) $c \mid \gcd(a, b)$.
15. A composite number n for which $a^n \equiv a \pmod{n}$ is called :
- (A) A Pseudoprime. (B) A Prime.
 (C) A pseudoprime to the base a . (D) An absolute pseudoprime.

16. The composite numbers n that are pseudoprime to every base a are called :

- (A) Pseudoprime. (B) Prime.
(C) Pseudoprime to the base a . (D) Absolute pseudoprime.

17. If $p, q_1, q_2, \dots, q_n$ are all primes and $p \mid q_1 q_2 \dots q_n$, then :

- (A) $p = q_k$ for some k . (B) $p = 2$.
(C) $q_k = 2$ for some k . (D) $p \mid q_k$ for some k .

18. If a is a solution of $P(x) \equiv 0 \pmod{n}$ and $a \equiv b \pmod{n}$, then :

- (A) ab is also a solution. (B) $a + b$ is also a solution.
(C) $a - b$ is also a solution. (D) b is also a solution.

19. If a is an odd integer, then $a^2 \equiv \text{---} \pmod{8}$.

- (A) 1. (B) 2.
(C) 3. (D) 4.

20. The solution of $25x \equiv 15 \pmod{29}$ is :

- (A) $x \equiv 18 \pmod{29}$. (B) $x \equiv 29 \pmod{29}$.
(C) $x \equiv 18 \pmod{19}$. (D) $x \equiv 17 \pmod{19}$.

FIRST SEMESTER (C.B.C.S.S.) DEGREE EXAMINATION, NOVEMBER 2020

Mathematics

MTS 1B 01—BASIC LOGIC AND NUMBER THEORY

(2020 Admissions)

Time : Two Hours and a Half

Maximum : 80 Marks

Section A

*Answer at least ten questions.**Each question carries 3 marks.**All questions can be attended.**Overall Ceiling 30.*

1. Find the converse and the contra positive of the implication "If today is Thursday, then I have a "test today."
2. What is the truth value of $\exists x P(x)$ where $P(x)$ is the statement " $x^2 > 10$ " and the universe of discourse consists of the positive integers not exceeding 4 ?
3. State which rule of inference is the basis of the following arguments: "It is below freezing now. Therefore, it is either below freezing or raining now".
4. Prove the theorem "The integer n is odd if and only if n^2 is odd".
5. Prove that there is a prime number greater than 3.
6. State Well-Ordering Property of $\mathbb{N}$.
7. Prove that there is no polynomial $f(n)$ with integral coefficients that will produce primes for all integers n .
8. If $d = (a, b)$ and d' is any common divisor of a and b , then prove that $d' \mid d$.
9. State Dirichlet's Theorem.
10. Find least number which when divided by 9 gives the remainder 8, when divided by 8 gives the remainder 7, when divided by 7 gives the remainder 6, ..., when divided by 3 gives the remainder 2, when divided by 2 gives the remainder 1.
11. Determine whether the linear Diophantine equations :
 $6x + 8y + 12z = 10$ and $6x + 12y + 15z = 10$ are solvable.
12. Let m and n be positive integers such that $m \mid n$. Then prove that $2^m - 1 \mid 2^n - 1$.

Turn over

13. Define $\tau(n)$ and $\sigma(n)$. Find $\tau(12)$ and $\sigma(12)$.
14. Prove that $\phi(n) = n - 1$ if and only if n is prime.
15. Solve the linear congruence $35x \equiv 47 \pmod{24}$.

(10 × 3 = 30 marks)

Section B

*Answer at least five questions.
Each question carries 6 marks.
All questions can be attended.
Overall Ceiling 30.*

16. Write the truth table for the implication $p \rightarrow q$.
17. Verify that $p \wedge T = p$ and $p \vee F = p$.
18. State and prove the Pigeonhole principle.
19. Find the number of positive integers ≤ 2076 and divisible by neither 4 nor 5.
20. State and prove Fermat's Little Theorem.
21. Find all solutions of the congruence $35x \equiv 47 \pmod{24}$.
22. Show that a positive integer a is self-invertible modulo p if and only if $a \equiv \pm 1 \pmod{p}$.
23. Show that $n^7 - n$ is divisible by 42.

(5 × 6 = 30 marks)

Section C (Essay Questions)

*Answer any two questions.
Each question carries 10 marks.*

24. (a) Show that $\sqrt[3]{3}$ is irrational.
(b) Express the statement
"Every student in this class has studied calculus" as a universal quantification.
25. (a) Explain the Two Queens Puzzle.
(b) Let a and b be any positive integers. Then prove that the number of positive integers $\leq a$ and divisible by b is $\lfloor a/b \rfloor$.
26. (a) Using Euclidean Algorithm calculate $\gcd(12378, 3054)$. Also represent the greatest common divisor as a linear combination of 12378 and 3054.
(b) State Fundamental Theorem of Arithmetic and find the canonical decomposition of 2520.
27. (a) State and prove Wilson's Theorem.
(b) Prove that $5^{2n+2} - 24n - 25$ is divisible by 576.

(2 × 10 = 20 marks)

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MTS 1B 01—BASIC LOGIC AND NUMBER THEORY

(Multiple Choice Questions for SDE Candidates)

1. The truth value of given statement is $4 + 3 = 7$ or 5 is not prime :
- (A) False. (B) True.
2. $(A \vee F) \vee (A \vee T)$ is always :
- (A) True. (B) False.
3. The number of primes is :
- (A) Finite. (B) Infinite.
(C) Uncountable.
4. Difference of two distinct prime numbers is :
- (A) Odd and prime. (B) Even and composite.
(C) None of the mentioned.
5. For $n \geq 1$, there are at least _____ primes less than 2^2 .
- (A) n . (B) $n - 1$.
(C) $n + 1$.
6. If $a \equiv b \pmod{n}$, then :
- (A) $n \mid a$ and $n \mid b$. (B) $n \mid b$ only.
(C) $n \mid (a - b)$.
7. $a \equiv b \pmod{n}$ and $b \equiv c \pmod{n}$, then :
- (A) $a \equiv c \pmod{n}$. (B) $a = b$.
(C) $a = c$.
8. If $a \equiv b \pmod{n}$, then : .
- (A) $a - b = n$. (B) $a - b = kn$, for some integer k .
(C) $a + b = kn$, for some integer k .

9. If $a \equiv b \pmod{n}$, and $m \mid n$, then :
- (A) $a \equiv b \pmod{.}$ (B) $a6 \equiv b \pmod{m}.$ /
- (C) $a \equiv b \pmod{(n/m)}.$
10. If $a \equiv b \pmod{n}$ and $\gcd(a, n) = d$, then $\gcd(b, n)$ is :
- (A) $a.$ (B) $d.$
- (C) $nd.$
11. A positive integer is divisible by 9 if and only if the sum of the digits in its decimal representation is divisible by :
- (A) 3. (B) 81.
- (C) 9.
12. If $\gcd(a, n) = 1$, then the congruence $ax \equiv b \pmod{n}$ has :
- (A) Infinitely many solutions modulo n .
- (B) Unique solution modulo n .
- (C) More than one solution modulo n .
13. The system of linear congruences $ax + by \equiv r \pmod{n}$ and $cx + dy \equiv s \pmod{n}$ has a unique solution modulo n whenever :
- (A) $\gcd(ad - bc, n) = 1.$ (B) $\gcd(ad, bc) = 1.$
- (C) $\gcd(ad, bc) = n.$
14. If p is a prime, then for any integer a :
- (A) $a^p \equiv a \pmod{p}.$ (B) $a^{p-1} \equiv 1 \pmod{p}.$
- (C) $a^{p-1} \equiv -1 \pmod{p}.$
15. $\sigma(12)$ is :
- (A) 28. (B) 27.
- (C) 16.

16. $\sigma(n) = n + 1$ if and only if :
- (A) n is an odd number. (B) n is an even number.
(C) n is a prime number.
17. The functions τ and σ are both multiplicative functions. The statement is :
- (A) False. (B) True.
(C) Partially true.
18. Which of the following statement is true ?
- (A) The functions τ and σ are both multiplicative functions.
(B) The Euler's Phi-function is injective.
(C) The Euler's Phi-function is not Multiplicative.
19. Given integers a, b, c , $\gcd(a, bc) = 1$ if and only if :
- (A) $\gcd(a, b) = 1$ and $\gcd(a, c) = 1$. (B) $\gcd(a, b) = 1$ and $\gcd(b, c) = 1$.
(C) $\gcd(a, b) = 1$ and $\gcd(a, c) = b$.
20. $\phi(2^3)$ is :
- (A) 2. (B) 3.
(C) 7.

FIRST SEMESTER (CBCSS-UG) DEGREE EXAMINATION, NOVEMBER 2021

Mathematics

MTS 1B 01—BASIC LOGIC AND NUMBER THEORY

(2020 Admissions)

Time : Two Hours and a Half

Maximum : 80 Marks

Section A

*(Answer any number of questions.**Each carries 2 marks.**Maximum marks that can be earned from this section is 25)*

1. What is a logical statement or a proposition ?
2. Define a vacuously true statement and give an example.
3. If $a \equiv b \pmod{n}$ and $c \equiv d \pmod{n}$, show that $ac \equiv bd \pmod{n}$.
4. Prove or disprove : Every composite number n has a prime factor $\leq \lfloor \sqrt{n} \rfloor$.
5. If $(a, b) = d$, show that a/d and b/d are relatively prime.
6. State Dirichlet's Theorem.
7. If n is a positive integer such that $(n-1)! \equiv -1 \pmod{n}$, show that n is a prime.
8. State Euler's theorem.
9. State the Division algorithm for integers.
10. What are linear diophantine equations in two variables x and y .
11. State the necessary and sufficient condition(s) for a linear congruence $ax \equiv b \pmod{m}$ to have a unique solution.
12. Use the divisibility criterion to check whether 10000234 divisible by eight or not.
13. Express the g.c.d (100,13) as a linear combination of 100 and 13.

Turn over

14. Compute the value of $\varphi(100)$.
15. Explain how canonical decomposition is useful in finding the least common multiple of two positive integers.

Section B

(Answer any number questions from this section.

Each question carries 5 marks.

Maximum that can be earned from this section is 35)

16. Evaluate the boolean expression $\sim [(x \leq y) \wedge (y > z)]$ when $x = 3, y = 4$ and $z = 5$.
17. Construct the truth table of $(p \rightarrow q) \wedge (q \rightarrow p)$.
18. Prove that there is no positive integer between 0 and 1.
19. There are n guests at a party. Each person shakes hands with everybody else exactly once. Define recursively the number of handshakes $h(n)$ made.
20. Prove that no prime of the form $4n + 3$ can be expressed as the sum of two squares.
21. Compute the remainder when 3^{247} is divided by 25.
22. Solve the linear congruence $35x \equiv 47 \pmod{24}$.
23. Let b be an integer ≥ 2 . Suppose $b + 1$ integers are randomly selected. Prove that the difference of two of them is divisible by b .

Section C

(Answer any two questions from this section.

Each question carries 10 marks.

Maximum that can be earned from this section is 20)

24. (a) Construct truth table for $p \rightarrow q \leftrightarrow \sim p \wedge q$
- (b) Prove by the method of contradiction : The square of an odd integer is odd. Moreover, rewrite the proposition symbolically with $UD = \text{set of all integers}$.

25. (a) State and prove the weak version of principle of mathematical induction.
- (b) Define the least common multiple $[a, b]$ of two positive integers and show that $[a, b] = \frac{ab}{(a, b)}$.
26. (a) Find the remainder when 24^{1947} is divided by 17.
- (b) State and prove the Fermat's little theorem.
27. State and prove the Fundamental theorem of arithmetic.

**FIRST SEMESTER (CBCSS—UG) DEGREE EXAMINATION
NOVEMBER 2021**

Mathematics

MTS 1B 01—BASIC LOGIC AND NUMBER THEORY

(2019 Admissions)

(Multiple Choice Questions for SDE Candidates)

Time : 15 Minutes

Total No. of Questions : 20

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MTS 1B 01—BASIC LOGIC AND NUMBER THEORY

(Multiple Choice Questions for SDE Candidates)

1. What is the value of x after this statement, assuming initial value of x is 5 ? 'If x equals to one then $x = x + 2$ else $x = 0$.
(A) 1. (B) 3.
(C) 0. (D) 2.
2. The compound statement $A \rightarrow (A \rightarrow B)$ is false, then the truth values of A, B are respectively :
(A) T, T. (B) F, T.
(C) T, F. (D) F, F.
3. Let $P(x)$ denote the statement " $x > 7$ ". Which of these have truth value true ?
(A) $P(0)$. (B) $P(4)$.
(C) $P(6)$. (D) $P(9)$.
4. When to proof $P \rightarrow Q$ true, we proof P false, that type of proof is known as :
(A) Direct proof. (B) Contrapositive proofs.
(C) Vacuous proof. (D) Mathematical Induction.
5. In the principle of mathematical induction, which of the following steps is mandatory :
(A) Induction hypothesis. (B) Inductive reference.
(C) Induction set assumption. (D) Minimal set representation.
6. For any positive integer m ,... is divisible by 4.
(A) $5m^2 + 2$. (B) $3m + 1$.
(C) $m^2 + 3$. (D) $m^3 + 3m$.
7. Suppose that $P(n)$ is a propositional function. Determine for which positive integers n the statement $P(n)$ must be true if : $P(1)$ is true ; for all positive integers n , if $P(n)$ is true then $P(n + 2)$ is true :
(A) $P(3)$. (B) $P(2)$.
(C) $P(4)$. (D) $P(6)$.

8. Let $\gcd(a, b) = d$. If c divides a and c divides b , then :
- (A) $c \leq d$. (B) $c \geq d$.
 (C) $c = 1$. (D) Cannot determine c with the given information.
9. If a and b are non-zero integers with $a \mid b$, then $\gcd(a, b)$ equals :
- (A) $|a|$. (B) b .
 (C) ab . (D) a .
10. The product of any three consecutive integers is divisible by :
- (A) 36. (B) 9.
 (C) 6. (D) 8.
11. If a is an odd integer then $\gcd(3a, 3a + 2)$ equals :
- (A) 3. (B) 5.
 (C) 1. (D) 2.
12. If a, b are two distinct prime number than highest common factor of a, b is :
- (A) 2. (B) 0.
 (C) 1. (D) ab .
13. $(1001111)_2 = \dots$
- (A) 79. (B) 89.
 (C) 69. (D) 99.
14. What is the one's complement of the number 1010110 ?
- (A) 1111111. (B) 0101001.
 (C) 1100110. (D) None of the mentioned.
15. An integer n is called a pseudoprime if :
- (A) $n \mid 2^n - 2$. (B) n is composite and $n \mid 2^n - 2$.
 (C) n is prime and $n \mid 2^n - 2$. (D) n is composite and $n \mid 2^n - 1$.

16. A composite number n for which $a^n \equiv a \pmod{n}$ is called :
- (A) A pseudoprime. (B) A prime.
(C) A pseudoprime to the base a . (D) An absolute pseudoprime
17. The composite numbers n that are pseudoprime to every base a are called :
- (A) Pseudoprime. (B) Prime.
(C) Pseudoprime to the base a . (D) Absolute pseudoprimes.
18. If $p, q_1, q_2, \dots, q_n$ are all primes and $p \mid q_1 q_2 \dots q_n$, then :
- (A) $p = q_k$ for some k . (B) $p = 2$.
(C) $q_k = 2$ for some k . (D) $p \mid q_k$ for some k .
19. If a is a solution of $P(x) \equiv 0 \pmod{n}$ and $a \equiv b \pmod{n}$, then :
- (A) ab is also a solution. (B) $a + b$ is also a solution.
(C) $a - b$ is also a solution. (D) b is also a solution.
20. The Sieve of Eratosthenes is used for finding :
- (A) All primes below a given integer.
(B) All even numbers below a given integer.
(C) All odd numbers below a given integer.
(D) All composite numbers below a given integer.

**FIRST SEMESTER (CBCSS—UG) DEGREE EXAMINATION
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Mathematics

MTS 1B 01—BASIC LOGIC AND NUMBER THEORY

(2019 Admissions)

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Maximum : 80 Marks

Section A

Answer any number of questions.

Each question carries 2 marks.

Maximum 25 marks.

1. What is meant by a *proposition* ? Is " $3 + 4 = 9$ " a proposition ? Why ?
2. What is meant by *negation* of a proposition ? Write the negation of the proposition "Ravi is taller than Joseph".
3. What is meant by the *contraposition* of an implication ? Give the contrapositive implication of "If $a > b$, then $b < c$."
4. Prove that every nonempty set of nonnegative integers has a least element.
5. Express $(231)_{10}$ in base 3.
6. Determine whether 1601 is a prime number.
7. Express $(28, 12)$ as a linear combination of 28 and 12.
8. Define the LCM of two numbers a, b . What is the LCM of 22 and 33 ?
9. Define the relation a congruent to b modulo n . Is 23 congruent to 4 modulo 5? Why?
10. Define complete set of residues modulo m . Give an example for such a set modulo 7 starting at 15.
11. Find the remainder when 16^{53} is divided by 7.
12. State Fermat's Little theorem and Wilson's theorem.
13. Find the remainder when 245^{1040} is divided by 18.

Turn over

14. Define *multiplicative* functions. Show that $f(n) = n^2$ is multiplicative.
15. Define the function σ . Evaluate $\sigma(12), \sigma(18)$.

Section B

Answer any number of questions.

Each question carries 5 marks.

Maximum 35 marks.

16. Prove directly that the product of any two odd integers is an odd integer and that the product of any two even integers is an even integer.
17. Let b be an integer $b \geq 2$. Suppose $b + 1$ integers are randomly selected. Prove that the difference of atleast two of them is divisible by b .
18. Prove that every integer $n \geq 2$ has a prime factor.
19. Define Fermat numbers. Show that $641 | f_5$.
20. Find the general solution to the LDE $1492x + 1066y = -4$.
21. Let a and b be positive integers. Prove that $[a, b] \times (a, b) = ab$.
22. Prove the Wilson's theorem : If p is a prime, then $(p - 1)! \equiv -1 \pmod{p}$. Without using it, verify that $10! \equiv -1 \pmod{11}$.
23. Prove that if f is a multiplicative function, then $F(n) = \sum_{d|n} f(d)$ is also multiplicative.

Section C

Answer any two questions.

Each question carries 10 marks.

Maximum 20 marks.

24. (a) Prove by contradiction : There is no largest prime number.
- (b) Prove by using contrapositive : "If the square of an integer is odd, then the integer is odd".
- (c) There is a male barber in a certain town. He shaves all those men and only those men who do not shave themselves. Explain why the question "does the barber shave himself?" results in a paradox.

25. (a) State the division algorithm and prove it.
- (b) Find the quotient q and remainder r when 305 is divided by 16.
- (c) State and prove the pigeonhole principle.
26. If a cock is worth five coins, a hen three coins, and three chicks together one coin, how many cocks, hens, and chicks, totaling 100, can be bought for 100 coins ?
27. (a) Let p be a prime and a any integer such that $p \nmid a$. Prove that the least residues of the integers $a, 2a, 3a, \dots, (p-1)a$ modulo p are a permutation of the integers $1, 2, 3, \dots, (p-1)$.
- (b) Let m be a positive integer and a any integer with $(a, m) = 1$. Prove that $a^{\phi(m)} \equiv 1 \pmod{m}$.

19. Find the area of the region between the x -axis and the graph of $f(x) = x^3 - x^2 - 2x$, $-1 \leq x \leq 2$.

(5 × 5 = 25 marks)

Section C

Answer any **one** question.

The question carries 11 marks.

20. (a) Find the area of the region in the first quadrant that is bounded above by $y = \sqrt{x}$ and below by the x -axis and the line $y = x - 2$.

(b) Find $\frac{dy}{dx}$ if $y = \int_1^{x^2} \cos t \, dt$.

21. (a) Find the absolute extrema of $h(x) = x^{2/3}$ on $[-2, 3]$.

- (b) Find the volume of the solid generated by the revolution about the x -axis of the loop of the

curve $y^2 = x^2 \frac{3a - x}{a + x}$.

(c) Evaluate $\lim_{x \rightarrow 0} \left(\frac{1}{x^2} - \frac{1}{\sin^2 x} \right)$.

(1 × 11 = 11 marks)

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(C) T, F. (D) F, F.
3. $(A \vee F) \vee (A \vee T)$ is always :
(A) True. (B) False.
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(A) $P(0)$. (B) $P(4)$.
(C) $P(6)$. (D) $P(9)$.
5. When to proof $P \rightarrow Q$ true, we proof P false, that type of proof is known as :
(A) Direct proof (B) Contrapositive proofs.
(C) Vacuous proof. (D) Mathematical Induction.
6. In the principle of mathematical induction, which of the following steps is mandatory ?
(A) Induction hypothesis. (B) Inductive reference.
(C) Induction set assumption. (D) Minimal set representation.
7. For any positive integer $m, \dots$ is divisible by 4.
(A) $5m^2 + 2$. (B) $3m + 11$.
(C) $m^2 + 3$. (D) $m^3 + 3m$.
8. Let $\gcd(a, b) = d$. If c divides a and c divides b , then :
(A) $c \leq d$. (B) $c \geq d$.
(C) $c = 1$. (D) Cannot determine c with the given information.

9. If a and b are relatively prime then :
- (A) $a \mid b$. (B) $b \mid a$.
 (C) $\gcd(a, b) = 1$. (D) $\text{lcm}(a, b) = 1$.
10. The product of any three consecutive integers is divisible by :
- (A) 36. (B) 9.
 (C) 6. (D) 8.
11. If a is an odd integer then $\gcd(3a, 3a + 2)$ equals :
- (A) 3. (B) 5.
 (C) 1. (D) 2.
12. If a, b are integers such that $a > b$ then $\text{lcm}(a, b)$ lies in :
- (A) $a > \text{lcm}(a, b) > b$. (B) $a > b > \text{lcm}(a, b)$.
 (C) $\text{lcm}(a, b) \geq a > b$. (D) None of the mentioned.
13. $(1001111)_2 = \text{---}$
- (A) 79. (B) 89.
 (C) 69. (D) 99.
14. The linear Diophantine equation $ax + by = c$ has a solution if and only if :
- (A) $\gcd(a, c) \mid b$. (B) $\gcd(a, b) \mid c$.
 (C) $\gcd(c, b) \mid a$. (D) $c \mid \gcd(a, b)$.
15. A composite number n for which $a^n \equiv a \pmod{n}$ is called :
- (A) A pseudoprime. (B) A prime.
 (C) A pseudoprime to the base a . (D) An absolute pseudoprime.
16. The composite numbers n that are pseudoprime to every base a are called :
- (A) A pseudoprime. (B) A prime.
 (C) A pseudoprime to the base a . (D) An absolute pseudoprime.
17. If a is a solution of $P(x) \equiv 0 \pmod{n}$ and $a \equiv b \pmod{n}$, then :
- (A) ab is also a solution. (B) $a + b$ is also a solution.
 (C) $a - b$ is also a solution. (D) b is also a solution.

18. If a is an odd integer then $a^2 - 1$ is :

- (A) A multiple of 7. (B) A multiple of 8.
(C) A multiple of 11. (D) A multiple of 9.

19. If a is an odd integer, then $a^2 \equiv \text{---} \pmod{8}$:

- (A) 1. (B) 2.
(C) 3. (D) 4.

20. The solution of $25x \equiv 15 \pmod{29}$ is:

- (A) $x \equiv 18 \pmod{29}$. (B) $x \equiv 29 \pmod{29}$.
(C) $x \equiv 18 \pmod{19}$. (D) $x \equiv 17 \pmod{19}$.

**FIRST SEMESTER (CBCSS-UG) DEGREE EXAMINATION
NOVEMBER 2021**

Mathematics

MTS 1B 01—BASIC LOGIC AND NUMBER THEORY

(2021 Admissions)

Time : Two Hour and a Half

Maximum : 80 Marks

Section A

Answer atleast ten questions.

Each question carries 3 marks.

All questions can be attended.

Overall ceiling 30.

1. Verify that $p \vee p \equiv p$ and $p \wedge p \equiv p$.
2. Let $P(x)$ denote the statement " $x > 3$." What is the truth value of the quantification $\exists x P(x)$, where the universe of discourse is the set of real numbers ?
3. State the barber paradox presented by Bertrand Russell in 1918.
4. Prove that if n is a positive integer, then n is odd if and only if $5n + 6$ is odd.
5. Prove the following formula for the sum of the terms in a "geometric progression" :

$$1 + r + r^2 + \dots + r^n = \frac{1 - r^{n+1}}{1 - r}.$$

6. Let a and b positive integers such that $a | b$ and $b | a$. Then prove that $a = b$.
7. Briefly explain Mahavira's puzzle.
8. Find the number of positive integers ≤ 2076 and divisible by neither 4 nor 5.
9. Prove that every composite number n has a prime factor $\leq \lfloor \sqrt{n} \rfloor$.
10. Show that any two consecutive Fibonacci numbers are relatively prime.

Turn over

11. Let a and b be integers, not both zero. Then prove that a and b are relatively prime if and only if there exist integers α and β such that $1 = \alpha a + \beta b$.
12. Prove that if $a \mid c$ and $b \mid c$, and $(a, b) = 1$, then $ab \mid c$.
13. Prove that every integer $n \geq 2$ has a prime factor.
14. Let f_n denote the n^{th} Fermat number. Then prove that $f_n = f_{n-1}^2 - 2f_{n-1} + 2$, where $n \geq 1$.
15. Express $\gcd(28, 12)$ as a linear combination of 28 and 12.

(10 × 3 = 30 marks)

Section B

Answer atleast five questions.

Each question carries 6 marks.

All questions can be attended.

Overall ceiling 30.

16. Show that the propositions $p \vee (q \wedge r)$ and $(p \vee q) \wedge (p \vee r)$ are logically equivalent.
17. Show that the assertion "All primes are odd" is false.
18. Let b be an integer ≥ 2 . Suppose $b + 1$ integers are randomly selected. Prove that the difference of two of them is divisible by b .
19. If p is a prime and $p \mid a_1 a_2 \dots a_n$, then prove that $p \mid a_i$ at for some i , where $1 \leq i \leq n$.
20. Show that $11 \times 14n + 1$ is a composite number.
21. There are infinitely many primes of the form $4n + 3$.
22. Show that $2^{11213} - 1$ is not divisible by 11.
23. Prove that if $n \geq 1$ and $\gcd(a, n) = 1$, then $a^{\phi(n)} \equiv 1 \pmod{n}$.

(5 × 6 = 30 marks)